

Construct validity of multi-attribute language-model features: a pre-deployment screen and the cost of omitting it

Abstract

Analytics teams increasingly obtain several conceptually distinct scores per document from a single language-model prompt and treat them as separate features. The marginal cost of requesting one more attribute has fallen to almost nothing; the risk that the extra attribute is not a separate measurement has not. This paper argues that a construct-validity screen belongs at the front of such a pipeline, specifies one that requires no labelled data and no additional model calls, and measures what omitting it costs. The screen names an external criterion for each attribute in advance and asks whether the attribute tracks its own criterion *conditional on* the others. We apply it to a three-attribute decomposition of financial news into polarity, novelty and materiality, combined multiplicatively so that any near-zero factor suppresses the signal, a design we also characterise axiomatically: continuity, strict monotonicity and ratio-scale separability force the combiner into the Cobb–Douglas family, whose exponents the axioms leave free. Across 134,037 headlines for 35 US-listed companies over 2010-01-05 to 2023-12-29, the screen fails. Novelty and materiality correlate at 0.871; only materiality tracks an external criterion, predicting absolute market-adjusted returns at $t = 12.70$, while novelty adds nothing to mechanical staleness beyond what materiality already supplies and enters with the wrong sign. Two of three attributes are one latent judgement reported twice. The downstream programme confirms every implication: freely estimated exponents place the materiality elasticity at zero, the gated aggregate predicts no better than plain, count-weighted or threshold-filtered polarity, and adding it to a walk-forward selective forecaster moves precision at fixed coverage by between -1.2 and $+0.4$ percentage points with every interval covering zero. That programme required 408 fitted objects; the screen required 3, and its verdict was available before the first of them.

Keywords: Construct validity, Large language models, Feature engineering, Data quality, Text analytics, Selective prediction

1. Introduction

An instruction-tuned language model will answer as many questions about a document as a prompt cares to ask. One call can return the sentiment, the urgency and the credibility of a support ticket; the severity, the novelty and the confidence of a clinical note; or the tone, the freshness and the importance of a news item. Each answer arrives on the scale requested, in the format requested, and at a marginal cost close to zero. For an analytics function under pressure to enrich its feature inventory, this is an unusually cheap source of supply.

It is cheap in the wrong place. The cost that fell is the cost of *asking*; the cost of being wrong about what came back did not move. A prompt that requests k attributes obtains k columns, but there is no mechanism that guarantees k measurements. A model may reduce several requested attributes to one internal judgement and report it k times under k labels, and the output gives no sign of it: the columns vary, they correlate imperfectly rather than perfectly, they respond sensibly to worked examples, and, if the calls are memoised, they reproduce exactly. Every internal consistency check a data team would ordinarily run is passed. What fails is only the assumption that the columns are about different things — and that assumption is precisely what the downstream feature design rests on.

This is a data-quality problem of a kind that the analytics literature is not yet equipped for. Conventional data quality asks whether values are complete, timely, consistent and accurate against a source of record (Wang and Strong, 1996; Batini et al., 2009). A prompted attribute has no source of record. Its quality question is not whether the number is correct but whether the *construct* it names exists as a separate thing in the instrument that produced it. Psychometrics has asked exactly that question for seventy years, under the heading of convergent and discriminant validity (Campbell and Fiske, 1959; Cronbach and Meehl, 1955), and has an answer that transfers.

1.1. What this paper does

We propose that a construct-validity screen be run *before* any feature is built from a multi-attribute prompt, and specify one that a team can

afford to run routinely. For each attribute, name in advance an observable criterion that the attribute must track if it measures what its label claims, and that the other attributes have no particular reason to track. Then test each attribute against its own criterion *conditional on* the others. The conditioning is the whole design: when one latent judgement drives every attribute, each attribute shows a respectable marginal association with every criterion, and only the conditional test exposes that none of them contributes anything the others do not.

We then measure what skipping the screen costs, on a case we can report without charity because we designed it, believed in it and built it. The case is a three-attribute decomposition of financial news — polarity, novelty and materiality — combined as a product, so that a stale or immaterial item is suppressed however extreme its tone. We ran the screen and the full downstream modelling programme, in that order, on 134,037 headlines covering 35 US-listed companies. The screen fails, and every subsequent result is an unpacking of the same failure.

The managerial version of the finding is a ratio. The downstream programme comprised 408 fitted objects: 21 predictive regressions, 12 nested horse-race coefficients, an exponent grid of 81 points at each of 3 horizons, 27 robustness re-estimations and 105 walk-forward selective forecasters. The screen comprised 3: two criterion regressions and one decile profile, on artefacts the study already had on disk. It required 0 additional calls to the scoring model. Its verdict was available before any of the 408 were run, and it implied all of them.

1.2. Contributions

1. **A pre-deployment validity screen for multi-attribute language-model features** (Section 4.3), stated as a five-step procedure with the conditional test at its centre, together with the three practical properties — no labels, no extra model calls, no fitted model — that make it affordable enough to be mandatory rather than aspirational.
2. **Evidence that the failure mode is real, with a worked diagnosis** (Section 5.3). Novelty and materiality correlate at 0.871; under the screen, materiality predicts absolute market-adjusted returns at $t = 12.70$ while novelty adds nothing to the staleness criterion that materiality does not already supply, and enters absolute returns with the opposite sign. The three-attribute decomposition is empirically a two-attribute one.

3. **A measurement of the omission cost** (Sections 5.4–5.7 and Table 2), which is what turns the screen from good practice into a decision with a number attached.
4. **A characterisation of the design under test** (Section 3). We prove a veto bound (Proposition 1) and show (Proposition 2) that continuity, strict monotonicity, sign fidelity and ratio-scale separability force the combiner into the Cobb–Douglas family $\text{sign}(s)|s|^{\alpha_0}\nu^\alpha\mu^\beta$. The axioms do not select the exponents, which is what makes the design falsifiable rather than definitional, and Section 5.6 estimates them.
5. **Deployment evidence on calibration and abstention** (Section 5.8) from an append-only ledger of 621 resolved interval forecasts issued by a production system, including a diagnosed and repaired calibration failure. It concerns the decision stage of the pipeline and is kept strictly apart from the claims about text.

Section 2 places the work; Section 3 states the design under test; Section 4 covers data, the screen and the downstream programme; Section 5 reports results; Section 6 draws the managerial implications; and Section 7 concludes.

2. Related work

2.1. Data quality, and the part of it that prompts break

Information-quality research has long treated fitness for use as multi-dimensional, spanning accuracy, completeness, timeliness, consistency and interpretability (Wang and Strong, 1996; Batini et al., 2009), and data-management practice has operationalised those dimensions as rules evaluated against a reference. Feature stores and machine-learning platforms inherit that framing: a feature is monitored for drift, missingness and range violations (Polyzotis et al., 2018; Breck et al., 2017). All of these presume that what the column *denotes* is settled and only its values are in question.

A prompted attribute inverts the presumption. Its values may be perfectly complete, perfectly timely and perfectly reproducible while the column denotes something other than its name. No drift monitor detects this, because nothing drifts; no range check detects it, because the range is exactly the one requested. The relevant tradition is instead the psychometric one, in which a measurement instrument earns the right to a construct label by demonstrating convergent validity with criteria that construct should track and discriminant validity against constructs it should not (Campbell and Fiske, 1959; Cronbach and

Meehl, 1955). Our screen is a compact, one-shot adaptation of that idea to the case where the instrument is a prompt.

2.2. *Language models as measurement instruments*

Instruction-tuned models return structured numeric judgements from prompts written as rubrics (Brown et al., 2020), and applied fields have adopted them quickly, in finance for return prediction (Lopez-Lira and Tang, 2023) and for reading corporate policy (Jha et al., 2024). Their self-reported confidence is imperfectly calibrated and is perceived by users as better than it is (Steyvers et al., 2025), which motivates a division of labour we adopt throughout: the model is never asked to forecast, only to *decompose* an item along defined axes, and forecasting and uncertainty quantification are left to a separate stage that can be calibrated against outcomes. The reliability of such judgements — whether two runs, or two models, agree — is increasingly reported. Their *validity* is not, and reliability does not imply it: an instrument can be perfectly consistent and consistently wrong.

2.3. *Relevance and novelty are established constructs*

We make no claim to have discovered the attributes we test. Commercial news analytics have scored relevance and event novelty for well over a decade — vendor feeds routinely attach a relevance score, an event novelty score and an event sentiment score to each item, and a large empirical literature filters on those fields before running its tests. On the academic side the novelty axis has a canonical antecedent in Tetlock (2011), who separates genuinely new stories from reprints of stale information and finds that investors react to the stale component; Boudoukh et al. (2019) show that identifying which news is *relevant* materially changes the measured information content of a news flow; Dang et al. (2015) study commonality in news flow across markets; and Kelly et al. (2021) treat which text gets written at all as an object of study. Hillert et al. (2014) document how coverage intensity interacts with momentum, and DellaVigna and Pollet (2009) show that identical information moves prices differently depending on when attention is available to receive it.

What is new here is not the attributes but the question asked of them: whether a general-purpose model, prompted zero-shot under a published rubric, in fact delivers them as *separate* measurements, and what a team loses by assuming it does.

2.4. Text, tone and outcomes

The application domain has a long empirical literature. Tetlock (2007) linked pessimistic media tone to downward price pressure and subsequent reversal, and Tetlock et al. (2008) showed that negative language in firm-specific news predicts fundamentals and returns. Antweiler and Frank (2004) studied message boards; Loughran and McDonald (2011) demonstrated that domain-specific word lists matter because general-purpose ones systematically mislabel domain terms, developed further in their survey (Loughran and McDonald, 2016); and Garcia (2013) showed the effect of tone is state-dependent. Supervised and neural text models improve on dictionaries (Ke et al., 2019; Huang et al., 2023), and machine learning is now standard in empirical asset pricing (Gu et al., 2020). Across this literature the per-item output is a scalar tone; freshness and importance, when handled at all, are handled by ad hoc pre-filters. That convention is the baseline our design set out to beat, and Section 5.9 finds it hard to beat.

2.5. Calibration and selective prediction

Modern classifiers are frequently miscalibrated (Guo et al., 2017); the standard remedies are Platt scaling (Platt, 1999) and isotonic regression (Zadrozny and Elkan, 2002), assessed by expected calibration error and the Brier score (Brier, 1950). Stacking (Wolpert, 1992) complicates this by creating more than one score distribution, a failure mode we diagnose and repair in Section 4.6. Abstention is an old idea (Chow, 1970) with a modern theory (Geifman and El-Yaniv, 2017; El-Yaniv and Wiener, 2010) and a distribution-free relative in conformal prediction (Angelopoulos and Bates, 2023). Applied machine learning in this domain remains dominated by accuracy and area-under-the-curve reporting, which is fragile under backtest overfitting (Bailey et al., 2017) and multiple testing (Harvey et al., 2016; Romano and Wolf, 2005), and comparatively little of it reports whether the probabilities can be trusted.

3. Theory: the design under test

3.1. Attributes and the combined signal

Let e denote a textual event — in our data, a news headline — concerning an entity, here a listed equity. The prompt requests three attributes:

- polarity $s(e) \in [-1, 1]$, the sign and size of the implied value change;

- novelty $\nu(e) \in [0, 1]$, the surprise relative to what is already known;
- materiality $\mu(e) \in [0, 1]$, the sensitivity of fundamental value to the event, independent of sign and of freshness.

The *event signal* is their product,

$$a(e) = s(e) \nu(e) \mu(e) \in [-1, 1]. \quad (1)$$

Writing *relevance* as $r(e) = \nu(e) \mu(e)$ gives $a(e) = s(e) r(e)$. The reading is a first-order decomposition of an expected move into a direction, a sensitivity and a surprise magnitude. Additive tone scoring retains only the first factor, which is why it responds to items that are loud but stale or cosmetic.

3.2. Aggregation

Events $\mathcal{E}$ touching one entity in one period are combined by a relevance-weighted mean over items that clear a materiality floor μ_0 :

$$A = \frac{\sum_{e \in \mathcal{E}: \mu(e) \geq \mu_0} r(e) a(e)}{\sum_{e \in \mathcal{E}: \mu(e) \geq \mu_0} r(e)} = \frac{\sum r(e)^2 s(e)}{\sum r(e)}. \quad (2)$$

Relevance therefore enters twice — once inside the event signal and once as the weight — so a novel and material item dominates quadratically while routine chatter is suppressed. We set $\mu_0 = 0.15$, the value used by the production system; Section 5.9 sweeps it.

3.3. Two properties of the product form

Proposition 1 (Veto bound). *For every event e ,*

$$|a(e)| \leq \min\{|s(e)|, \nu(e), \mu(e)\}.$$

In particular $a(e) = 0$ whenever any single factor is zero, and the aggregate of Eq. (2) satisfies $|A| \leq \max_e |s(e)|$.

Proof. Each factor has modulus at most one and $\nu, \mu \geq 0$, so $|a(e)| = |s(e)| \nu(e) \mu(e)$ is a product of three numbers of modulus at most one, which cannot exceed the modulus of any one of them; this gives the per-event bound and the zero case. For the aggregate,

$$|A| \leq \frac{\sum_e r(e)^2 |s(e)|}{\sum_e r(e)} \leq \max_e |s(e)| \cdot \frac{\sum_e r(e)^2}{\sum_e r(e)} \leq \max_e |s(e)|,$$

the last step using $\sum_e r(e)^2 \leq (\max_e r(e)) \sum_e r(e) \leq \sum_e r(e)$ because every $r(e) \in [0, 1]$. The weights are non-negative but sub-stochastic, so the aggregate contracts polarities toward zero and never amplifies them. $\square$

Proposition 1 is the formal content of “a single weak factor vetoes the event”: a stale ($\nu \rightarrow 0$) or immaterial ($\mu \rightarrow 0$) item is capped near zero whatever its tone. No additive combination has this property, since a weighted sum $\alpha s + \beta \nu + \gamma \mu$ is generally nonzero when one term vanishes.

Why a product rather than some other veto-respecting form? An earlier version of this argument assumed that scaling any one attribute by λ scales the combiner by λ , which is close to assuming the conclusion. The following statement replaces that assumption with the standard conditions of conjoint measurement (Debreu, 1960; Gorman, 1968; Krantz et al., 1971) together with the observation that each attribute is measured on a ratio scale.

Proposition 2 (Representation). *Let $f : [-1, 1] \times (0, 1] \times (0, 1] \rightarrow [-1, 1]$ satisfy*

- (A0) normalisation: $f(1, 1, 1) = 1$;
- (A1) sign fidelity: $\text{sign } f(s, \nu, \mu) = \text{sign}(s)$;
- (A2) veto: $f = 0$ whenever any of $|s|, \nu, \mu$ equals 0;
- (A3) regularity: f is continuous and, on the interior, $|f|$ is strictly increasing in each of $|s|, \nu, \mu$;
- (A4') ratio-scale separability: for each argument there is a function h of one variable such that multiplying that argument by $\lambda \in (0, 1]$ multiplies $|f|$ by $h(\lambda)$, whatever the values of the other two arguments.

Then there exist exponents $\alpha_0, \alpha, \beta > 0$ with

$$f(s, \nu, \mu) = \text{sign}(s) |s|^{\alpha_0} \nu^\alpha \mu^\beta. \quad (3)$$

Proof. Work on the positive orthant $|s|, \nu, \mu \in (0, 1]$ and write $F = |f|$. Fix the second and third arguments. By (A4') there is h_0 with $F(\lambda x, \nu, \mu) = h_0(\lambda)F(x, \nu, \mu)$ for all admissible λ, x . Applying this twice,

$$\begin{aligned} h_0(\lambda_1 \lambda_2)F(x, \nu, \mu) &= F(\lambda_1 \lambda_2 x, \nu, \mu) = h_0(\lambda_1)F(\lambda_2 x, \nu, \mu) \\ &= h_0(\lambda_1)h_0(\lambda_2)F(x, \nu, \mu), \end{aligned}$$

and since $F > 0$ on the interior, h_0 satisfies the multiplicative Cauchy equation $h_0(\lambda_1\lambda_2) = h_0(\lambda_1)h_0(\lambda_2)$. By (A3) h_0 is continuous and positive, so its only solutions are the power functions $h_0(\lambda) = \lambda^{\alpha_0}$ (Aczél, 1966), with $\alpha_0 > 0$ because F is strictly increasing in its first argument. Setting $x = 1$ gives $F(\lambda, \nu, \mu) = \lambda^{\alpha_0}F(1, \nu, \mu)$. Repeating for the second and third arguments yields $\alpha, \beta > 0$ and $F(|s|, \nu, \mu) = |s|^{\alpha_0}\nu^\alpha\mu^\beta F(1, 1, 1)$, and (A0) fixes $F(1, 1, 1) = 1$. Restoring the sign by (A1) gives Eq. (3). Continuity extends the representation to the closed domain, where (A2) is automatic because all three exponents are strictly positive. $\square$

Remark 1. Proposition 2 delivers the *family*, not a member of it. Eq. (1) is the special case $\alpha_0 = \alpha = \beta = 1$, and nothing in (A0)–(A4′) selects it. This is deliberate. The axioms assert that the three attributes act on separate multiplicative scales and cannot compensate for one another, which is the substantive claim; the elasticities with which they act are an empirical matter, and Section 5.6 estimates α and β freely rather than assuming H_0 : $\alpha = \beta = 1$. Proposition 1 holds for every member of the family in the weaker form $|f| \leq \min\{|s|^{\alpha_0}, \nu^\alpha, \mu^\beta\}$, so the veto survives whatever the exponents turn out to be.

The design in Eq. (1)–(2) has one silent prerequisite, which the rest of this paper is about. A product of three gates behaves like a product of three gates only if there *are* three gates. If two of the attributes are one attribute wearing two labels, the second gate cannot bind where the first does not, the exponents are not separately identified, and the design degenerates to a two-attribute one without any warning appearing in the output.

4. Material and methods

4.1. Two bodies of evidence, kept apart

The study draws on two datasets that answer different questions and are never used to support claims about each other (Table 1).

A large historical news panel (US equities)... Testing the decomposition needs a long panel with dense per-item news and clean prices. We use FNSPID (Dong et al., 2024), a public corpus of time-stamped financial news headlines keyed to tickers. From the symbols with usable price history we take the best-covered names and remove exchange-traded funds, for which “the sensitivity of this firm’s fundamental value to the event” has no clean meaning; the

exclusion list is enumerated in the panel-building script rather than pattern-matched, so it is auditable. That leaves 35 operating companies. Prices are split- and dividend-adjusted daily bars; returns are market-adjusted using per-symbol betas estimated on training windows only, and a Parkinson range estimator (Parkinson, 1980) supplies a volatility proxy.

A production forecast ledger (Indian large caps).. The forecasting system of Sections 4.6 and 4.7 is deployed on liquid National Stock Exchange of India large-capitalisation equities. Since April 2026 it has written every issued forecast to an append-only ledger — entity, timestamp, target date, point forecast, interval, nominal confidence, directional probability and horizon — with each row later stamped with the realised price and outcome. Rows are scored at first look and never retro-edited. This ledger supplies the interval-coverage and calibration evidence of Section 5.8. It cannot speak to the value of the text decomposition: it stores no per-event return labels and its history is far too short to fit anything.

We state the resulting limitation at the outset rather than in a closing caveat. *The validation market is not the deployment market.* No historical Indian headline corpus of comparable density is available to us, so “does the decomposition carry information?” is answered on US data and “does a calibrated selective forecaster behave as designed in production?” is answered on Indian data.

Table 1: The two bodies of evidence. The validation panel answers whether the decomposition carries information; the deployment ledger answers whether a calibrated selective forecaster behaves as designed in production. They are drawn from different markets and are never used to support claims about each other.

	Validation panel	Deployment ledger
Market	US listed equities	NSE India large caps
Source	FNSPID headline corpus	Live append-only forecast ledger
Entities	35 operating companies	7 names (live), 54 (walk-forward)
Span	2010-01-05 to 2023-12-29	19 Apr 2026 to 12 Jun 2026
Sessions	3,521	—
Symbol-sessions	117,312	—
... with scored news	48,261	—
Headlines scored	134,037	—
Resolved forecasts	—	621
Question answered	Does gating carry information?	Is the deployed system honest?

4.2. Scoring the attributes

The attributes are produced by an instruction-tuned large language model prompted as a quantitative analyst and constrained to return a JSON array of $\{\nu, \mu, s\}$ triples, one per headline, in input order. Three design choices make the measurement reproducible.

First, the prompt fixes *numerical anchors* on every axis. For novelty, 0.0 is an exact restatement of information already priced or a generic market list, 0.2 an in-line and widely expected result, 0.6 an unexpected but modest development, and 1.0 a sudden shock such as a surprise regulatory action. For materiality, 0.0 is boilerplate or a price-move round-up, 0.4 a contract or ruling worth a few percent of revenue, 0.8 a transformative deal or guidance reset, and 1.0 an existential event. For polarity the anchors run from -1.0 (a fraud allegation) through 0.0 (a neutral reshuffle) to $+1.0$ (a large earnings beat or a takeover at a premium). Anchoring matters more here than in ordinary tone scoring, because the product in Eq. (1) amplifies disagreement on any single axis.

Second, every scored item is memoised in a content-addressed store keyed by the SHA-1 hash of the entity and the normalised headline, so repeated runs are deterministic and the scoring history is auditable.

Third, the attributes are scored *zero-shot*: no labelled training data, no fine-tuning, and no market outcome enters the prompt, so the scores cannot encode look-ahead information about the returns they are later used to predict.

The production Indian system uses one commercial instruction-tuned model for this role; the US validation panel was scored with a different vendor’s model. Using different models in the two settings is not ideal for comparability, but it supplies evidence on a question that matters for the screen’s portability — whether the anchored rubric is model-specific — and Section 5.1 reports test–retest agreement across the two.

4.3. The validity screen

This is the procedure the paper argues should run first. It is an adaptation of convergent and discriminant validity (Campbell and Fiske, 1959; Cronbach and Meehl, 1955) to the case where the instrument is a prompt, compressed to what a data team can execute in an afternoon.

1. **Name a criterion per attribute, in advance.** For each attribute, state an observable quantity that it must track *if* it measures what its

label claims, and that the other attributes have no particular reason to track. The criterion must be computable without the attribute itself. Here, novelty is assigned *mechanical staleness* — lexical overlap with the same entity’s recent coverage — and materiality is assigned *outcome magnitude*, the absolute market-adjusted return of the session.

2. **Test conditionally, not marginally.** Regress each criterion on all attributes jointly, or equivalently examine partial correlations. This step carries the diagnostic weight. If one latent judgement drives every attribute, each will show a respectable *marginal* association with every criterion, and only the conditional test reveals that none adds anything to the others.
3. **Inspect the shape before summarising it.** A rank correlation is meaningless if the relationship is not monotone. Plot each attribute against deciles of its criterion first, and only then report a coefficient.
4. **Control the mechanical confounds of the criterion.** A criterion computed from a comparison set inherits that set’s properties: a document with few predecessors scores as novel by construction. The size of the comparison set enters as a control.
5. **Check that the criterion is the right *kind* of quantity for the downstream task.** An attribute can be valid and still useless. One that legitimately measures magnitude cannot sharpen a directional forecast. This step converts a validity result into a design decision.

Three properties make the screen cheap enough to be routine. Neither criterion required labelled data. Neither required additional model calls: mechanical staleness is computed from the corpus already collected and outcome magnitude from prices already needed for the study, so the screen adds 0 scoring calls to a panel of 134,037. And nothing in it requires a predictive model to be fitted, so it precedes rather than competes with the modelling programme. Note also that the outcome criterion validates the *instrument* and is never an input to any signal: checking a measurement against an outcome is legitimate, whereas building the measurement from that outcome would not be.

4.4. *The downstream programme, and what it cost*

To measure the omission cost we ran the modelling programme that a team skipping the screen would run. Every aggregator below is built from

the same scored events, so no comparison is confounded by differences in the underlying text.

Mean polarity $\bar{s}$, the standard additive recipe.

Threshold-filtered polarity the mean polarity of events clearing the materiality floor. This is the baseline that matters most, because it is what practitioners actually do with a vendor relevance score.

Count-weighted polarity $\sum_e s(e)/\sqrt{|\mathcal{E}|}$, which rewards repeated coverage as a naive “sum the sentiment” pipeline implicitly does.

Additive combiner the mean of $(s+\nu+\mu)/3$: the natural non-multiplicative way to use exactly the same three inputs.

Single-gate ablations the same relevance-weighted aggregate built from $s\nu$ alone and from $s\mu$ alone, isolating each gate’s contribution.

On top of these we estimated the exponents of Eq. (3) on a grid of 81 points at each horizon, ran a nested horse race, swept the analyst’s three consequential choices as robustness, and fitted a walk-forward selective forecaster with and without the gated signal. Table 2 counts the resulting objects against the screen. The point of the table is not that 408 fits are expensive in machine time — they are not — but that each carries analyst attention, review time and an opportunity for a specification choice to be made in the direction the analyst hopes for. The screen has three such objects and none of them is a predictive model.

4.5. *Timing convention*

Every headline carries a UTC publication stamp. It is converted to America/New_York and assigned to the session whose 16:00 ET close first follows it, so the aggregate for session d contains only information public before that close and is used to predict returns realised from session $d + 1$ onward. The convention is applied uniformly and is the single most important guard in the study: a related analysis in earlier work by the present authors produced a spuriously significant result that turned out to be a time-zone join artefact, and this convention was adopted after that diagnosis.

Table 2: What the screen costs and what it pre-empts. The upper panel is the validity screen of Section 4.3; the lower panel is the modelling programme reported in Section 5, whose every conclusion the screen anticipates. Counts are of fitted objects, derived from the stored result files rather than asserted. Neither criterion in the upper panel required a labelled example or an additional call to the scoring model.

Stage	Unit counted	Count
<i>Panel A: the validity screen</i>		
Criterion regressions	one per attribute	2
Decile profiles inspected	one per criterion pair	1
Additional scoring-model calls	headlines rescored	0
Total fitted objects		3
<i>Panel B: the downstream programme</i>		
Predictive regressions	aggregator $\times$ horizon	21
Nested horse-race coefficients	term $\times$ horizon	12
Exponent-grid evaluations	81 per horizon, 3 horizons	243
Robustness re-estimations	variation $\times$ setting $\times$ horizon	27
Walk-forward selective forecasters	5 variants $\times$ 7 years $\times$ 3 horizons	105
Total fitted objects		408

4.6. From gated signal to calibrated probability

The aggregate A is one feature among several entering a forecaster that also sees price, momentum, trend, reversal and volatility features, and that emits per name and horizon a point forecast, a price interval and a directional probability $\hat{p}$. We treat the forecaster as a black box and study its calibration, measured by the expected calibration error over B equal-width bins,

$$\text{ECE} = \sum_{b=1}^B \frac{|G_b|}{N} \left| \bar{y}_b - \bar{p}_b \right|, \quad (4)$$

with $\bar{p}_b$ and $\bar{y}_b$ the mean predicted probability and mean realised outcome in bin b , and by the Brier score $N^{-1} \sum_i (\hat{p}_i - y_i)^2$ (Brier, 1950).

The first deployed version of this model reported next-day up-probabilities clustered near 0.80 against a realised accuracy near 0.51. The cause was a stacking domain mismatch (Wolpert, 1992): an isotonic calibrator had been fitted on the out-of-fold *average of the base learners* and was then applied to the *stacked* prediction, a different distribution, and the blend itself was never calibrated. The repair is a three-stage procedure — refit the isotonic map on the base-average it was meant to correct, form the blend, then fit a final Platt scaler (Platt, 1999) on a held-out fold and apply it to test and live

rows. The transferable lesson is that a calibrator is bound to the distribution it was fitted on, and stacking silently creates more than one.

4.7. The conviction gate

Calibration makes $\hat{p}$ honest, not skilful. If short-horizon direction is close to unpredictable, a calibrated probability sits near the base rate almost everywhere and acting on every name is pointless. The remedy mirrors the text-stage veto: act only when conviction clears a threshold. For a horizon of H sessions the system emits an UP call when $\hat{p} \geq \tau^*$ and abstains otherwise, with τ^* chosen on training data as the smallest τ whose fired bucket reaches a target precision while still firing at least a minimum fraction of the time,

$$\tau^* = \min\{\tau : \text{prec}(\tau) \geq \pi_0 \text{ and } \phi(\tau) \geq \phi_0\}, \quad (5)$$

where $\text{prec}(\tau) = \Pr[y = 1 \mid \hat{p} \geq \tau]$ and $\phi(\tau) = \Pr[\hat{p} \geq \tau]$, with $\pi_0 = 0.60$ and $\phi_0 = 0.05$. The gate is one-sided because confident DOWN calls are destroyed by the upward drift of equities, so the system emits long-or-neutral signals only.

Reporting a single operating point invites the objection that τ^* was tuned to the number being reported. We therefore also report the full *risk-coverage curve* and its area (AURC): sorting predictions by confidence and sweeping the threshold traces the selective risk accepted at each coverage level, and the area under that curve summarises the whole trade-off without reference to any threshold (El-Yaniv and Wiener, 2010; Geifman and El-Yaniv, 2017).

4.8. Evaluation protocol and inference

All panel results are walk-forward by calendar year: for test year Y the model sees only sessions strictly before 1 January Y , the last 20% of the training sessions are held out to fit the probability calibrator and to choose τ^* , and neither is ever selected on test data. Test years run from 2014 to 2020. Inference respects the two dependence structures of a return panel: a market-wide shock hits every name on the same date, and returns are persistent within a name. We therefore report two-way clustered standard errors by date and symbol (Cameron et al., 2011), cross-checked against Fama–MacBeth cross-sectional slopes with Newey–West correction (Fama and MacBeth, 1973; Newey and West, 1987). Multi-session horizons are evaluated on non-overlapping windows as the primary specification, and confidence intervals for differences between models come from a block bootstrap that resamples whole dates, preserving cross-sectional correlation.

5. Results

We report in the order a sceptical reader would want. First, does the instrument measure anything stable? Second, are the three attributes the separate things Section 3 assumes? Third, does the gated aggregate predict at all? And last, does it predict better than the simpler recipes it is meant to replace. The answers are, in order: yes; *no*; yes; and no.

5.1. Reliability of the instrument

Scoring 134,037 headlines for 35 companies over 2010-01-05 to 2023-12-29 yields attributes with sensible central tendencies: mean novelty 0.189, mean materiality 0.157 and mean polarity 0.039, the last confirming the mild positive skew the news-tone literature reports. Two properties of the instrument deserve comment before anything is built on it.

First, the model reports on a *coarse grid* rather than a continuum: the mass sits on multiples of 0.1, visible as the discrete bars of Figure 1(a). This is a property of anchored prompting, not an artefact of our aggregation, and it means the attributes carry less resolution than their $[0, 1]$ range suggests.

Second, the scores are *deterministic by construction*, because every item is memoised under a hash of the entity and normalised headline. That is a property of the memoisation, not of the model, so we also rescored a random subsample of 250 headlines with the cache bypassed and a *different* vendor’s instruction-tuned model, which tests the stronger and more useful claim that the anchored rubric is portable rather than model-specific. Agreement is moderate and, tellingly, uneven across the attributes: the intraclass correlation (Shrout and Fleiss, 1979) is 0.779 for polarity, 0.727 for materiality and only 0.565 for novelty, with the event signal $a = s\nu\mu$ inheriting 0.601.

The ordering matters for how the rest of the paper reads. Polarity, the attribute with decades of instrument development behind it, is the one two independent models agree on most; novelty, the attribute on which the design’s distinctive claim rests, is the one they agree on least. Any true multiplicative effect is therefore being estimated through the noisiest available channel, and classical attenuation would bias the measured contribution of gating toward zero. We return to this in Section 6.5, because it is the most credible non-trivial explanation of the null results that follow. The subsample is small — 250 headlines, bounded by API quota rather than by design — so these coefficients are indicative rather than precise.

5.2. *The veto works; the attributes are not separable*

The gate does exactly what Proposition 1 promises. Table 3 shows that 56.3% of events fall below the materiality floor and are refused outright, and that 47.6% of symbol-sessions carrying news end with an aggregate of exactly zero. The contraction is equally visible: the standard deviation of the gated aggregate is 0.054 against 0.154 for mean polarity, so gating removes about two-thirds of the cross-sectional dispersion of the signal. Figure 1(c) is the per-session version of the same statement, with almost every point below the 45-degree line. On its own terms the mechanism is not in doubt.

The premise underneath it is. Multiplicative gating supposes that novelty and materiality are *separable* properties acting through different channels. In the measured data they are very nearly the same variable: the correlation between ν and μ across all 134,037 events is **0.871**, and Figure 1(b) shows the joint mass collapsing onto the diagonal. An item the model judges novel is almost always an item it judges material, and conversely.

This single fact anticipates every result that follows. If two of the three attributes carry nearly the same information, the second gate can add almost nothing once the first has acted, the product cannot separate itself from simpler combinations of the same inputs, and any attempt to estimate distinct elasticities for ν and μ will be poorly identified. All three happen.

Two very different things could produce that correlation. It may be a fact about the *world* — genuinely new firm news usually *is* the consequential news — or a fact about the *instrument*, an inability of the model to hold two related concepts apart under one prompt. The two have opposite implications for what a team should do next, so the screen exists to settle the question rather than leave it open.

5.3. *The screen: does each attribute track its own criterion?*

Each attribute has an external criterion it must track if it measures what it claims, and neither criterion required further model calls.

Novelty should track mechanical staleness. Following the stale-news literature (Tetlock, 2011), we measure how much a headline repeats what was already written about the same ticker: one minus the maximum TF-IDF cosine similarity to that ticker’s headlines over the preceding thirty days. The comparison set is strictly earlier in time, so the measure is causal and uses no returns. It is available for 99.9% of headlines (133,933 of 134,037), with a median of 38 prior documents.

Table 3: Descriptive statistics. Upper panel: the three axes and their products at the level of the individual headline. Lower panel: the session-level aggregators, computed on symbol-sessions carrying at least one scored headline. The final column is the share of exactly-zero values, which is the share of events or sessions the gate vetoes outright.

	Mean	SD	p25	Median	p75	Zeros (%)
<i>Panel A: per headline</i>						
Polarity s	0.039	0.207	0.000	0.000	0.200	52.0
Novelty ν	0.189	0.195	0.000	0.200	0.200	30.7
Materiality μ	0.157	0.166	0.000	0.100	0.200	34.4
Relevance $r = \nu\mu$	0.058	0.100	0.000	0.020	0.060	35.0
Event signal $a = s\nu\mu$	0.005	0.056	0.000	0.000	0.004	53.7
<i>Panel B: per symbol-session</i>						
Gated signal A	0.006	0.054	0.000	0.000	0.008	47.6
Relevance-filtered polarity	0.042	0.195	0.000	0.000	0.200	48.4
Mean polarity	0.038	0.154	0.000	0.000	0.100	36.8
Count-weighted polarity	0.058	0.227	0.000	0.000	0.200	36.8
Additive combiner	0.118	0.107	0.033	0.100	0.167	17.7
Novelty gate only ($s\nu$)	0.017	0.091	0.000	0.000	0.040	36.7
Materiality gate only ($s\mu$)	0.013	0.079	0.000	0.000	0.040	37.2

The correlation between novelty and materiality across all events is 0.871: the two gates are close to the same variable, which is the single fact behind most of what follows.

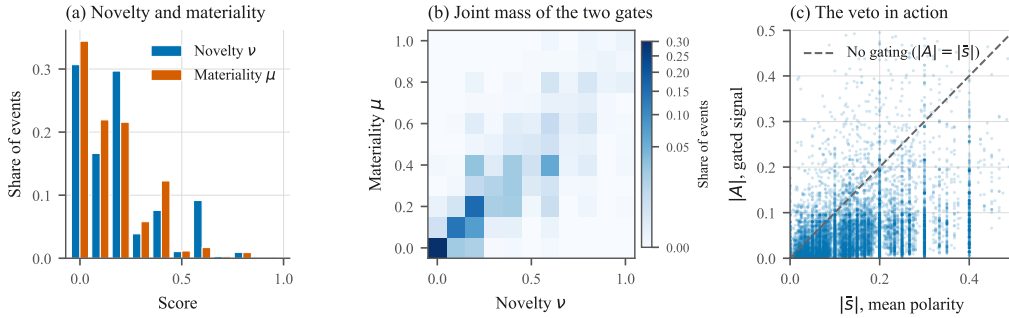


Figure 1: The three attributes and the veto they implement, measured on 134,037 scored headlines. (a) Marginal distributions of novelty and materiality; the model reports on a coarse grid, so the honest display is a discrete bar chart. (b) Joint mass of the two gates, on a power-law colour scale so the sparse high-novelty, high-materiality cells remain visible. (c) The gated session aggregate $|A|$ against the mean absolute polarity $|\bar{s}|$ of the same session: almost every point lies below the 45-degree line, which is Proposition 1 in the data.

Materiality should track outcome magnitude. Materiality is defined as the sensitivity of value to an event *regardless of direction*, so it should predict the absolute market-adjusted return of the session.

The screen separates the two attributes sharply, but not in the direction the design needs.

Novelty fails its own criterion.. Figure 2(a) plots both attributes against deciles of mechanical novelty, and the two curves are the same curve: parallel, identically shaped, separated by a constant. The relationship is also non-monotonic, rising from 0.167 at the stalest decile to 0.230 at decile 4 and then falling to 0.093 at the most lexically novel decile — the model assigns its *lowest* novelty to the headlines that share least vocabulary with recent coverage. Controlling for materiality and for the size of the comparison set, novelty’s standardised coefficient on staleness is -0.033 ($t = -6.51$) against materiality’s -0.022 ($t = -3.91$): distinguishable from zero, but essentially the same number, so novelty carries no staleness information that materiality does not carry equally. The partial correlations tell the same story — -0.014 for novelty given materiality, -0.014 for materiality given novelty — as does the dense subsample with at least twenty prior documents (-0.155 and -0.157 , identical to three decimals). Step 3 of the screen earns its place here: the raw rank correlation between the novelty attribute and mechanical novelty is -0.159 , which read alone would suggest a modest inverse relation rather than the hump the decile profile actually shows.

Materiality passes its own criterion, decisively.. Over 130,828 events, a one standard deviation increase in materiality is followed by 54.3 additional basis points of absolute market-adjusted return ($t = 12.70$, clustered by date), while novelty enters the same regression with the *opposite* sign at -26.3 basis points ($t = -8.69$). Whatever common factor drives the 0.871 correlation, the materiality label is the one that aligns with a real external quantity; conditional on it, the novelty label points the wrong way (Figure 2(b)).

The verdict is therefore *instrument*, and specifically the novelty attribute. The model is largely reporting a single latent “importance” judgement on two channels; the channel labelled materiality happens to align with event magnitude, and the channel labelled novelty does not measure novelty in the sense the design requires.

Step 5 of the screen then supplies the implication that matters most, and supplies it before any forecasting model exists. The surviving attribute,

materiality, predicts the *magnitude* of the move and not its *direction* — exactly what its definition promises. But Eq. (1) multiplies it into a *directional* signal. A quantity carrying information about how large a move will be cannot improve a forecast of which way it will go. That is the cleanest explanation for the materiality exponent estimating at zero in Section 5.6, and it was available from the screen alone.

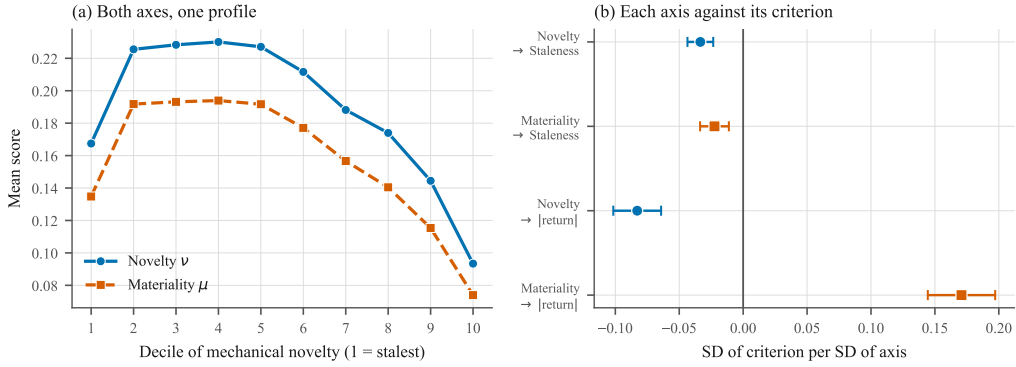


Figure 2: The validity screen. (a) Mean novelty and mean materiality across deciles of mechanical novelty: the two curves are parallel and identically shaped, which is what one variable measured twice looks like, and both are non-monotonic in mechanical novelty. (b) Standardised coefficients of each attribute on each criterion, in standard deviations of criterion per standard deviation of attribute, with 95% intervals from date-clustered standard errors. Materiality predicts absolute return strongly; novelty does not track mechanical staleness beyond what materiality already tracks.

5.4. Predictive content of the aggregators

The gated aggregate does predict returns. At the one-session horizon a one standard deviation move in A is followed by 12.2 basis points of market-adjusted return, with a two-way clustered t of 4.14 and a Fama–MacBeth t of 3.70; its rank information coefficient is 0.023, averaged over 2,548 trading days with a Newey–West t of 4.44. At five sessions the effect is 17.8 basis points ($t = 2.12$) on non-overlapping windows. By twenty-one sessions nothing survives: every aggregator has $|t| < 0.9$, which is what one expects of headline-level information at a monthly horizon.

So the text carries signal. The question the study exists to answer is whether the *product* carries more of it than the recipes it is meant to displace, and Table 4 says it does not. At one session, mean polarity earns a

higher t -statistic than the gated aggregate (5.52 against 4.14); count-weighted polarity produces a larger coefficient still (14.1 basis points) and the highest information-coefficient t -statistic of the seven (4.72); and threshold-filtered polarity — the recipe a practitioner would actually deploy — matches the product almost exactly (10.0 basis points). The information coefficients of the gated aggregate, threshold-filtered polarity and count-weighted polarity are 0.0230, 0.0230 and 0.0228: identical for any practical purpose. The one aggregator that clearly underperforms is the additive combiner $(s + \nu + \mu)/3$ at 3.3 basis points and 0.010, which is worth stating plainly. *Adding* the three attributes is a bad idea, so the argument against additive combination in Section 3 survives; it is the step from a filter to a product that does not pay.

Figure 3 makes the overlap visible. At the horizons where anything is estimable, the confidence intervals of all seven aggregators sit on top of one another.

Table 4: Predictive regressions of forward market-adjusted returns on each aggregator, one aggregator at a time. Regressors are standardised, so a coefficient reads as basis points of forward return per one standard deviation of signal. Standard errors are two-way clustered by date and symbol; horizons beyond one session use non-overlapping windows. Stars mark significance at the 10%, 5% and 1% levels.

Aggregator	$H = 1$		$H = 5$		$H = 21$	
	bps	t	bps	t	bps	t
Gated signal A	12.2***	4.14	17.8**	2.12	11.7	0.60
Relevance-filtered polarity	10.0***	5.11	12.5*	1.82	10.4	0.46
Mean polarity	9.3***	5.52	13.6**	2.03	18.5	0.87
Count-weighted polarity	14.1***	4.78	17.1**	2.29	14.3	0.62
Additive combiner	3.3***	3.82	3.9	0.71	-3.3	-0.16
Novelty gate only ($s\nu$)	11.1***	4.71	15.8**	2.14	7.3	0.38
Materiality gate only ($s\mu$)	11.9***	4.53	15.1*	1.89	10.8	0.60

Largest estimation sample: 48,260 symbol-sessions.

5.5. The nested horse race

Entering all four terms together (Table 5) was intended to be the sharp test: multiplicative gating predicts that the product loads and absorbs the lower-order terms. What happens instead is that nothing is individually identified. At one session the gated aggregate carries the largest coefficient (16.3 basis points) but only $t = 1.58$, while mean polarity falls to $t = 1.26$ and the two single-gate ablations take opposite signs. At five and twenty-one

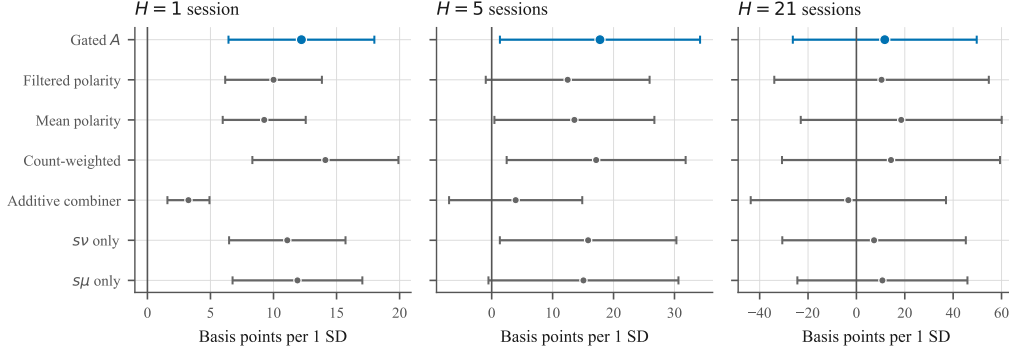


Figure 3: Forward-return coefficient on each aggregator, in basis points per one standard deviation of signal, with 95% intervals from two-way clustered standard errors. The gated signal is highlighted; the alternatives are shown in neutral ink.

sessions the coefficients grow large and alternate in sign — +50.3 for the product against −36.2 for the materiality gate at five sessions — with $|t| < 2$ throughout.

This pattern is diagnostic of collinearity, not of a hidden effect, and Section 5.2 already said why: with ν and μ correlated at 0.871 and all four regressors built from the same polarity, the design matrix is close to singular and the individual coefficients are not separately estimable. We therefore decline to read anything into the sign or size of any single term. The honest conclusion is a statement about what the data *cannot* answer: on this corpus the nested horse race lacks the power to distinguish the product from its components.

Table 5: The nested horse race. All four terms enter one regression, so each coefficient is the marginal contribution of that term given the others. Multiplicative gating predicts that the gated signal loads and absorbs the lower-order terms. Regressors standardised; two-way clustered t -statistics.

Term	$H = 1$		$H = 5$		$H = 21$	
	bps	t	bps	t	bps	t
Mean polarity	6.2	1.26	23.1*	1.92	85.5*	1.76
Novelty gate only ($s\nu$)	-9.1	-1.30	-15.2	-0.63	-103.8*	-1.77
Materiality gate only ($s\mu$)	0.4	0.05	-36.2	-1.25	-64.7	-0.59
Gated signal A	16.3	1.58	50.3	1.33	111.8	0.98

5.6. Free exponents

The horse race cannot separate the terms, but the exponent grid asks a cleaner question, because it varies the functional form directly instead of trying to identify collinear regressors. Remark 1 left α and β free; here we estimate them by sweeping $s\nu^\alpha\mu^\beta$ over a grid of 81 points on $[0, 2]^2$ and reading off the information coefficient of the resulting aggregate.

The answer is unambiguous and it is not the one the unit-exponent product predicts. At one session the grid maximum sits at $\alpha = 0.75$, $\beta = 0.00$ with information coefficient 0.0225: the materiality exponent is estimated at *exactly zero*, as Step 5 of the screen implied it would be. The unit-exponent product assumed by Eq. (1) scores 0.0210, and switching the gates off altogether — $\alpha = \beta = 0$, which collapses the aggregate to plain mean polarity — scores 0.0209. At five sessions the maximum *is* the no-gating corner, $\alpha = \beta = 0$ with 0.0156, against 0.0126 for the unit-exponent product: there, gating is not merely unhelpful but actively harmful.

Two conclusions follow. First, $H_0: \alpha = \beta = 1$ is rejected, and rejected in the direction of the null model rather than toward some richer weighting. Second, and more important for a team deciding where to spend effort, the entire surface spans only 0.0201 to 0.0225 in information coefficient (Figure 4). The objective is nearly flat in the exponents, so the choice of gating weights — including the choice not to gate — moves predictive accuracy by an amount small relative to its own sampling error. That flatness, rather than the location of any maximum, is the finding, and it is why every comparison in Section 5.7 lands inside its confidence interval.

5.7. Selective forecasting with and without the gated signal

This is the experiment the design has to pass: the conviction gate is fitted twice on identical walk-forward folds, once on price features alone and once on price features plus the gated text signal, and the reported quantity is the difference.

There is no difference. Adding A moves precision at a fixed 10% coverage by +0.37 percentage points at one session (95% CI [-1.26, +1.96], $p = 0.685$), -1.23 points at five sessions ([-3.52, +1.13], $p = 0.322$) and +0.01 points at twenty-one ([-0.74, +0.72], $p = 1.000$). Against threshold-filtered polarity rather than against price alone the gaps are +0.07, -0.83 and +0.37 points. Every interval contains zero, and the point estimates change sign across horizons. The risk-coverage curves of Figure 5 say the same thing without reference to any threshold: the curves for the four feature sets are

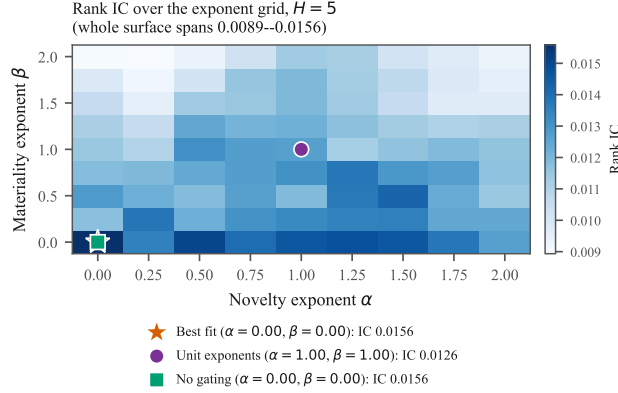


Figure 4: Information coefficient of the aggregate built from $s\nu^\alpha\mu^\beta$ over a grid of exponents. The unit-exponent product assumed by Eq. (1) is marked, as is the empirical maximum; Remark 1 explains why the axioms leave this an open question.

indistinguishable over the whole coverage range, and their AURCs differ in the third decimal.

One result in Table 6 does not fit the pattern, and we report it because it is the most interesting thing in the table. The *text-only* model — no price features at all — achieves the best AURC at both short horizons (0.479 against 0.485 for price-only at one session, and 0.468 against 0.473 at five) and the best precision at five sessions (55.3% against 54.0% for price-only, on a 53.3% base rate). The headlines therefore carry decision-relevant information that the price features do not, and a learner given the text alone finds it. What the learner cannot do is exploit that information *once price features are also available*: the combined model is no better than price alone. Whether this reflects genuine redundancy or a failure of the gradient-boosted learner to allocate capacity to a weak, sparse feature beside strong dense ones, we cannot determine from these data, and we would not want a reader to take the first explanation for granted.

5.8. Deployment evidence: coverage and calibration

The results above concern the text stage. The decision stage was studied on the production system, where the evidence is of a different kind: not whether a signal predicts, but whether a system in operation tells the truth about its own uncertainty.

It did not. Between 19 April and 12 June 2026 the system issued and

Table 6: Selective forecasting with and without the gated text signal. Precision is the realised up-rate of the most confident 10% of out-of-sample predictions, a fixed coverage that makes the variants comparable without reference to any tuned threshold; AURC is the area under the risk–coverage curve, for which lower is better. All figures are pooled over walk-forward test years. The lower panel bootstraps the precision gap by resampling whole dates.

Feature set	$H = 1$		$H = 5$		$H = 21$	
	Prec.	AURC	Prec.	AURC	Prec.	AURC
Price + gated signal A	51.5	0.479	52.4	0.472	59.7	0.458
Price + filtered polarity	51.4	0.482	53.8	0.472	59.1	0.460
Price + mean polarity	51.2	0.484	53.4	0.470	59.3	0.459
Price only	51.2	0.485	54.0	0.473	59.0	0.457
Text only	52.8	0.479	55.3	0.468	58.9	0.462
Always-up base rate	51.8%		53.3%		54.9%	

Precision gap at 10% coverage	Estimate (pp)	95% CI	p
$H = 1$, vs. price only	+0.37	[-1.26, +1.96]	0.685
$H = 1$, vs. filtered polarity	+0.07	[-1.70, +1.83]	0.938
$H = 1$, vs. mean polarity	-0.05	[-2.23, +2.14]	0.948
$H = 5$, vs. price only	-1.23	[-3.52, +1.13]	0.322
$H = 5$, vs. filtered polarity	-0.83	[-3.10, +1.40]	0.482
$H = 5$, vs. mean polarity	-0.65	[-2.96, +1.64]	0.587
$H = 21$, vs. price only	+0.01	[-0.74, +0.72]	1.000
$H = 21$, vs. filtered polarity	+0.37	[-0.23, +0.97]	0.252
$H = 21$, vs. mean polarity	+0.28	[-0.32, +0.84]	0.345

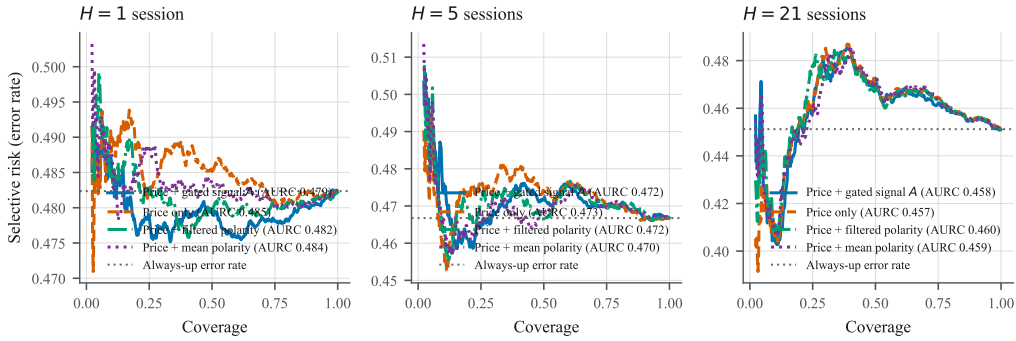


Figure 5: Risk–coverage curves for the walk-forward selective forecaster. Each curve traces the error rate accepted at every coverage level, so the comparison does not depend on the choice of any single conviction threshold; the area under each curve (AURC, lower is better) appears in the legend.

later resolved 621 price-interval forecasts on seven large caps, each carrying a nominal 90% interval with a median half-width of 5.6% of price. Had those intervals been calibrated, about 90% would have contained the realised price; empirical coverage was 70.7%, a 19.3 percentage-point shortfall (Table 7, Figure 6(a)). Most forecasts sat at the ten-day horizon, where coverage was 69.2% over 575 resolved forecasts. Nothing in the pipeline noticed, which is the point: overconfidence is silent unless somebody scores it.

The directional head failed in the same direction and for a diagnosable reason. It reported next-day up-probabilities clustered near 0.80 against a realised accuracy near 0.51, giving an expected calibration error of 0.345 and a Brier score above 0.366 — worse than predicting 0.5 every time. The cause was the stacking domain mismatch of Section 4.6, and the three-stage repair cut expected calibration error from 0.345 to 0.049 while leaving accuracy untouched. Calibration changed what the model *said about its confidence*, not what it predicted, which is precisely the distinction the design insists on.

With probabilities pulled back to the base rate, the walk-forward conviction gate on the 54-name deployment universe (2018–2026, 91,471 training rows and 50,272 out-of-sample rows) settles at $\tau^* = 0.63$ and fires on 5.6% of observations, abstaining on the rest. The fired bucket realises 60.6% directional precision against a 58.0% always-up base rate over 2,840 pooled calls: a 2.6 point edge with a 95% binomial interval of 58.8 to 62.4%. We do not over-read it. Per-year counts are small, the swing from +10.8 points in 2025 to −2.4 in 2024 is within sampling noise (Figure 6(b)), and the ranking area under the curve is about 0.47, so what edge exists lives in the high-conviction tail and not in the ordering. Crucially for this paper, *that gate contains no text feature at all*: it is evidence that calibrated abstention is worth building, not evidence for the decomposition.

5.9. Robustness

A null result is only worth reporting if it is not an artefact of the analyst’s choices, so we varied the three that matter, in 27 re-estimations.

The materiality floor. Sweeping μ_0 from 0 (no floor) to 0.40 leaves the one-session coefficient on A between 12.0 and 12.2 basis points with t between 3.97 and 4.14. The floor does not bind on the conclusion, which is what one expects when the exponent on materiality is estimated at zero. One detail cuts the other way and we note it: at five sessions the information coefficient rises from 0.013 at $\mu_0 = 0$ to 0.025 at $\mu_0 = 0.25$ and above, with the Newey–West t going from 1.24 to 2.25. A *hard threshold filter* thus helps where multiplicative

Table 7: Deployment evidence from the live forecast ledger. Every interval carries a nominal 90% level. Panel B reports the walk-forward conviction gate on the 54-name deployment universe, where precision is the realised up-rate of the fired high-conviction bucket and the base is the unconditional always-up rate.

<i>Panel A: interval coverage by horizon</i>				
Horizon	Resolved forecasts	Nominal	Empirical	
5 trading days	8	90.0%	100.0%	
10 trading days	575	90.0%	69.2%	
20 trading days	38	90.0%	86.8%	
All	621	90.0%	70.7%	

<i>Panel B: walk-forward conviction gate by test year</i>				
Test year	Fires	Fired precision	Always-up base	Edge (pp)
2022	365	61.6%	54.0%	+7.6
2023	290	68.3%	65.3%	+3.0
2024	932	53.3%	55.7%	−2.4
2025	822	69.0%	58.2%	+10.8
2026	431	54.5%	47.1%	+7.4
Pooled	2,840	60.6%	58.0%	+2.6

The pooled base is the unconditional always-up rate over all out-of-sample rows, not the fires-weighted mean of the yearly bases (55.9%); the pooled edge is therefore the more conservative of the two comparisons.

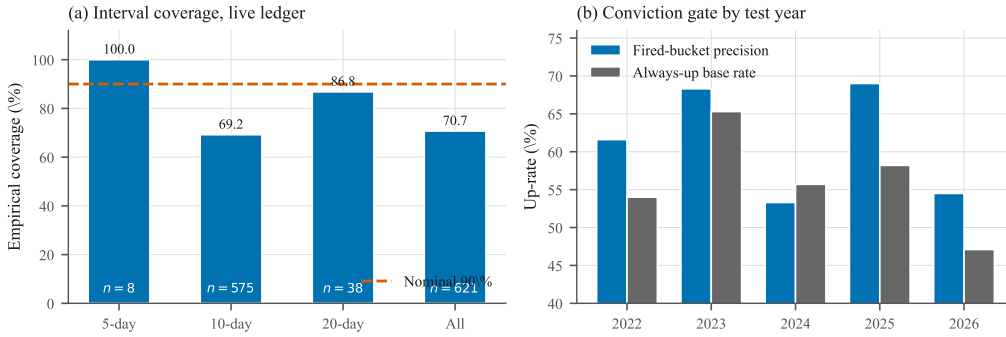


Figure 6: Deployment evidence. (a) Empirical coverage of nominal 90% price intervals on the live ledger, by horizon. (b) Fired-bucket precision of the conviction gate against the always-up base rate, by walk-forward test year, at roughly a 5.6% firing rate.

weighting does not — consistent with the exponent surface, and consistent with what practitioners already do with vendor relevance scores.

Overlapping versus non-overlapping windows. Using every session rather than every H -th weakens the five-session result from 17.8 basis points ($t = 2.12$) to 8.9 ($t = 1.68$), in the direction the overlapping-observation literature predicts. We report non-overlapping windows as the primary specification for that reason; neither choice changes the comparison between aggregators, which is what the study turns on.

Quiet sessions. Restricting the panel to symbol-sessions carrying scored news is the primary specification, since including quiet sessions pads the sample with rows where every text aggregator is identically zero and can only dilute any difference between them.

Repeatability of the instrument. Cross-model agreement is reported in Section 5.1. We regard it as the weakest link in the measurement chain and the first thing a replication should attack.

6. Discussion

6.1. Why the decomposition did not pay

Section 5.3 lets us be specific rather than speculative, and the answer has two parts.

One attribute is not measuring what it is named. Novelty fails its external criterion: it does not track mechanical staleness beyond what materiality already tracks, its profile against mechanical novelty is the same curve as materiality’s, and it is non-monotonic besides. Materiality passes decisively, predicting absolute market-adjusted returns at 12.70 standard errors. So the 0.871 correlation is not the world telling us that novel news is material news; it is one latent importance judgement reported twice under two labels, only one of which corresponds to something external. A product of two gates cannot behave like a two-gate product when only one gate exists.

The surviving attribute carries the wrong quantity for the task. Materiality is by definition about magnitude irrespective of sign, and that is exactly how it behaves. Multiplying it into a *signed* signal, as Eq. (1) does, asks a magnitude variable to sharpen a direction forecast, which it has no means of doing. This is the cleanest reading of the zero materiality exponent, and it points somewhere constructive: the same decomposition aimed at a volatility or absolute-return target, where materiality’s information is of the relevant kind, is a genuinely different experiment and one we have not run.

Attenuation remains a live secondary concern. Cross-model intraclass correlations are 0.779 for polarity but only 0.565 for novelty, and the product inherits 0.601; multiplying noisy factors compounds their noise, so classical errors-in-variables would bias the measured contribution of gating toward zero. We cannot rule this out, and it is a further reason the design deserves a rematch with a better instrument rather than a dismissal. What it does not license is treating the null as a measurement artefact by assumption — and in any case attenuation cannot explain why novelty points the *wrong way* on absolute returns, which is a validity failure and not a noise problem.

Note also what survived. The additive combiner performed worst of all the aggregators, so collapsing the attributes by addition is clearly wrong; and a *hard* materiality filter improved the five-session information coefficient from 0.013 to 0.025 where multiplicative weighting did not. Read together, these say that relevance information is worth using as a *filter* — decide which items to look at — but not as a continuous multiplicative *weight* on the ones that survive. The practitioner convention of thresholding a vendor relevance score, which we set out to improve on, appears to be the right call.

6.2. What the screen would have saved

The value of a screen is not that it produces a result but that it produces the result *early*, and Table 2 is the accounting. Every finding in Section 5 is an implication of the screen’s two regressions:

- *Novelty and materiality collapse into one attribute* $\Rightarrow$ the nested horse race cannot identify separate terms (Section 5.5), and the exponent surface is nearly flat (Section 5.6).
- *The surviving attribute measures magnitude, not direction* $\Rightarrow$ the materiality exponent estimates at zero against a signed target (Section 5.6).
- *Neither implication leaves room for an incremental gate* $\Rightarrow$ the selective forecaster with and without A differs by less than its own sampling error (Section 5.7).

The three screen objects and the 408 downstream objects are not comparable in machine time; a modern laptop runs both. They are comparable in the resource that actually binds an analytics team, which is the sequence of decisions an analyst makes while a result is still ambiguous. Each of the 243 exponent evaluations, 21 predictive regressions and 105 walk-forward

fits carries a specification choice — which horizon is primary, whether to report overlapping windows, which threshold defines coverage — and each such choice is an opportunity for a result to be nudged toward the one the analyst hoped for. Running the screen first removes the ambiguity that makes those choices consequential.

6.3. Managerial implications

For a team that is about to build on multi-attribute language-model scores, the findings translate into four specific rules.

1. **Treat every prompted attribute as an instrument requiring validation, not as a column requiring monitoring.** Existing feature-store controls — drift, missingness, range — are all passed by an attribute that measures the wrong construct. Add a validity gate at feature onboarding, and make an attribute’s criterion a mandatory field in its specification.
2. **Require the criterion to be named before the scores are generated.** A criterion chosen after inspecting the scores is a rationalisation. The version of this study that we would defend is the one in which each attribute’s criterion was fixed by its definition: novelty against staleness, materiality against outcome magnitude.
3. **Budget attributes by validated count, not requested count.** The marginal cost of an extra attribute in the prompt is near zero, but the marginal cost of an *unvalidated* extra attribute is the entire modelling programme built on it. In this study the ratio was 3 objects to 408.
4. **Use relevance as a filter, not as a weight.** Where the two are distinguishable in these data, the hard threshold wins: it lifted the five-session information coefficient from 0.013 to 0.025 where continuous multiplicative weighting moved nothing. This is a rare case of the simpler incumbent practice being vindicated against a more elaborate proposal.

There is also a negative recommendation. The most reusable part of the screen is step 5 — checking that the criterion is the right *kind* of quantity for the task — and it is the step most easily skipped, because it feels like philosophy rather than measurement. It was the step that predicted the single most consequential number in the study, the zero materiality exponent, and it costs nothing but a sentence of thought per attribute.

6.4. *Scope beyond this application*

Nothing in Section 3 is specific to equities. Any stream of timestamped textual events feeding a decision has the same structure, and the same failure mode when conceptually distinct attributes are requested from one prompt: triage of clinical alerts, operations tickets and content-moderation queues all ask how bad, how new and how consequential, and all are exposed to a model that answers the three questions with one judgement. The screen transfers directly, because its only requirement is that each attribute admit an observable criterion the others have no reason to track — for a ticket-urgency attribute, time-to-resolution; for a moderation-severity attribute, the eventual enforcement action.

The decision-stage half of the design is an instance of selective classification (Chow, 1970; Geifman and El-Yaniv, 2017), here paired with a probability whose calibration was repaired rather than assumed, and the interval-coverage failure of Section 5.8 is exactly the problem distribution-free methods (Angelopoulos and Bates, 2023) are built for.

6.5. *Limitations*

Five limitations bound what may be concluded, and we would rather state them than have them found.

The validation market is not the deployment market. The decomposition is tested on US headlines while the production system operates on Indian equities. We report no result that crosses that boundary, but a reader should not read the US panel as evidence about the NSE’s microstructure, disclosure regime or news supply, all of which differ.

The attributes are judgements, and one of them did not survive validation. Anchoring the prompt tightens the scores and Section 5.1 quantifies their repeatability, but repeatability is not validity: a scorer can be perfectly consistent and consistently wrong, which is close to what the screen finds for novelty. Every result about gating is therefore a joint test of the design and of this particular instrument, and a differently worded rubric, a larger model, or a two-stage prompt that scores the attributes independently might separate them where ours did not. We consider that the single most valuable follow-up.

The external criteria are proxies. Lexical similarity over a thirty-day window is a serviceable but crude stand-in for what the market already knew, and absolute return is a noisy realisation of an event’s true importance. A criterion failure is therefore weaker evidence than a criterion success:

materiality passing is strong evidence it measures something, whereas novelty failing is consistent with both a bad attribute and a bad proxy.

The comparison is against feasible baselines, not against a vendor feed. The most informative comparison would be against a commercial relevance and novelty product on the same headlines. We do not have a licence for one, so threshold-filtered polarity is our best available stand-in for what a practitioner would deploy.

Selection and multiple testing. We report several horizons, several aggregators and several feature sets. The risk-coverage curves and the non-overlapping-window specification remove the most common sources of optimism, and differences between models carry bootstrap intervals, but a reader who wants a single number with a family-wise guarantee should apply a stepdown correction (Romano and Wolf, 2005) to the horse-race column and read the result accordingly. Applied machine learning in this domain has a well-documented tendency to discover edges that do not survive (Bailey et al., 2017; Harvey et al., 2016), and we would rather our estimates be read with that prior in mind.

7. Conclusions

A textual event carries direction, surprise and consequence, and there is a clean argument that these should combine multiplicatively: the product makes freshness and relevance preconditions for influence, it caps the signal at its weakest factor, and under continuity, monotonicity and ratio-scale separability it is the form the attributes must take, up to exponents the axioms leave free.

Measured against real data, the argument does not pay. On 134,037 headlines covering 35 companies, the gated aggregate predicts market-adjusted returns at short horizons but no better than mean polarity, count-weighted polarity or threshold-filtered polarity; estimating the exponents freely puts materiality’s at zero and, at five sessions, collapses the aggregate to plain polarity; and adding the gated signal to a walk-forward selective forecaster moves precision at fixed coverage by between -1.2 and $+0.4$ percentage points, with every interval containing zero.

The explanation is in the measurement rather than in the market. Novelty and materiality, as an instruction-tuned model scores them, correlate at 0.871; validated against external criteria, only materiality survives, predicting absolute market-adjusted returns at 12.70 standard errors, while novelty fails

to track mechanical staleness beyond what materiality already tracks and enters absolute returns with the wrong sign. The three-attribute decomposition is empirically a two-attribute one, and the attribute that does measure something measures *magnitude*, which a signed product has no way to exploit.

Two things survive as positive findings. Adding the attributes is worse than multiplying them, so the case against additive combination stands. And using relevance as a hard filter rather than a continuous weight improves the five-session information coefficient from 0.013 to 0.025 — which is what practitioners already do with vendor relevance scores, now with evidence behind it. Separately, on the deployment side, nominal 90% intervals covered 70.7% of live outcomes, a three-stage recalibration cut expected calibration error from 0.345 to 0.049 without touching accuracy, and a conviction gate abstaining on 94% of names raised the thirty-day hit rate to 60.6% against a 58.0% base — evidence that calibrated abstention is worth building, independent of anything the text stage does.

The managerial conclusion is the ratio. The modelling programme that established all of the above comprised 408 fitted objects. The screen that implied all of it comprised 3, used no labelled data, and added 0 calls to the scoring model. As multi-attribute prompting spreads through analytics pipelines, the constraint on feature quality will not be the cost of generating attributes; it will be the discipline to check that the attributes generated are the ones that were asked for. Two experiments follow directly from this one and neither is run here: an instrument whose novelty channel passes an external validity check would give the design the fair test it has not yet had, and pointing the same decomposition at a volatility or absolute-return target asks the question the signed formulation cannot. The measurements reported here are designed to make both possible.

Declarations

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Declaration of generative AI in scientific writing. An instruction-tuned language model is used as a measurement instrument *within the system under study*, scoring headlines along the three attributes from the anchored rubric of Section 4.2; those outputs are persisted for audit. During manuscript preparation a large language model was used solely to improve language and readability. No generative tool produced the data, results, figures or interpretations, and the authors take full responsibility for the content of the publication.

Data availability. The FNSPID news corpus is publicly available (Dong et al., 2024); price data are public end-of-day records. The scored attribute values, the derived panel, the analysis code and the production forecast ledger underlying the reported tables and figures are available from the corresponding author on reasonable request.

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